From Games to Graphs: Categorical Composition of Genetic Algorithms Across Domains

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Categorical formalizations now exist for neural networks [6], reinforcement learning [10], compositional MDPs [3], and multi-agent systems [23]—yet evolutionary computation, one of the oldest families of optimization algorithms, remains outside this framework. At the same time, island-model practitioners have known for decades that migration topology affects search dynamics, but lack a mathematical theory explaining *why* ring outperforms star outperforms fully connected.

We bridge both gaps. Modeling GA operators as Kleisli morphisms over a population monad, we show that the island functor lifting strategies to multi-population systems is strict when islands evolve independently and lax when coupled by migration. The *laxator*—the natural transformation measuring the discrepancy—is the formal quantity governing diversity. A spectral bridge then connects the laxator's magnitude to the algebraic connectivity λ_2 of the migration graph, yielding a computable predictor: smaller λ_2 means less coupling and higher sustained diversity.

The prediction is confirmed across six unrelated domains—OneMax, maze generation, graph coloring, knapsack, checkers, and co-evolutionary card play (No Thanks!). The diversity ordering none $>$ ring $>$ star $>$ random $>$ fully connected holds with perfect rank correlation (Kendall's $W = 1.0$, $p = 0.00008$), with topology explaining $28.7\times$ more variance than domain. The spectral theory makes a further falsifiable prediction: at $n \geq 7$ islands, ring preserves strictly more diversity than star (reversing their n=5 relationship), confirmed with $p < 0.0001$. Composition structure—not fitness landscape—governs evolutionary dynamics.

1 Introduction

Evolutionary computation practitioners have long known that migration topology matters: island models with ring migration behave differently from those with fully connected migration, even on the same fitness landscape [21, 20]. Over the past three years, four independent groups have formalised a deeper structural insight: the way an optimisation pipeline *composes* its components determines dynamics more than the components themselves [6, 10, 3, 23]. Yet across all four programmes the same gap persists: none treats migration topology as a formal variable. There is no mathematical theory of *why* ring outperforms star outperforms fully connected.

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Main result. The diversity ordering none $>$ ring $>$ star $>$ random $>$ fully connected holds across six fundamentally different domains with perfect rank correlation (Kendall's $W = 1.0$, all 15 pairwise Spearman $\rho = 1.0$, $p = 0.00008$). A two-way ANOVA confirms topology explains $28.7\times$ more variance in diversity than domain. This ordering is a structural invariant, not an artefact of any fitness landscape—but explaining it requires all three parts of the argument to interlock.

Argument. *Category theory* identifies the key object. We model GA operators as Kleisli morphisms over a population monad (Section 2). The island functor lifting strategies to multi-population systems is strict when islands evolve independently and lax when coupled by migration. The *laxator*—the natural transformation measuring the discrepancy introduced by migration—is the formal quantity that governs diversity. *Graph theory* makes it computable. We connect laxator magnitude to the algebraic connectivity λ_2 of the migration graph (Section 3).[1] The spectral ordering $\lambda_2(\text{none}) = 0 < \lambda_2(\text{ring}) < \lambda_2(\text{star}) < \lambda_2(\text{FC})$ matches the diversity ordering and yields a falsifiable boundary prediction: at $n \geq 7$ islands $\lambda_2(C_n) < 1.0 < \lambda_2(K_{1,n-1})$, so ring and star cross over (reversing their n=5 relationship where $\lambda_2(C_5) = 1.382$). *Empirical validation* confirms both. Experiments across six unrelated domains—OneMax, maze generation, graph colouring, knapsack, checkers, and co-evolutionary card play—yield $W = 1.0$, and the n=7 spectral prediction holds with $p < 0.0001$ (Section 4).

Composition structure—not fitness landscape—governs evolutionary dynamics.

2 Categorical Framework

We formalize GA operators as Kleisli morphisms over a monad capturing the computational effects of evolution: configuration, logging, and randomness. Following Moggi [16], who showed that computational effects are uniformly captured by monads, our key observation is that *evolutionary dynamics are computational effects too*: stochastic mutation, stateful PRNG threading, and diversity-metric accumulation.

2.1 The Evolution Monad

Definition 1 (Population)**.** *Let Σ be a type of* genomes *(e.g., $\mathbb{R}^n$ for weight vectors, $\{0,1\}^k$ for bitstrings). A* scored individual *is a pair $(g, f) \in \Sigma \times \mathbb{R}$, where f is the fitness of genome g. A* population *is a finite list $P = [(g_1, f_1), \ldots, (g_n, f_n)]$ of scored individuals. We write $\mathsf{Pop}(\Sigma)$ for the type of populations over Σ.*

Definition 2 (Evolution monad)**.** *The* evolution monad *is the composite monad*

$$T \;=\; \mathsf{Reader}(\Gamma) \times \mathsf{Writer}(\Lambda) \times \mathsf{State}(S),$$

where Γ is the configuration context (population size, mutation rate, crossover rate, tournament size), Λ is a log monoid of diversity metrics, and S is the PRNG state. Concretely, a computation in T is a function $\Gamma \to S \to (A \times S \times \Lambda)$. The unit $\eta_A : A \to TA$ lifts a pure value with unchanged state and empty log. The multiplication $\mu_A : T^2A \to TA$ sequences computations by threading PRNG state and concatenating logs. Since Reader, Writer, and State compose via well-known distributive laws [16], the composite T inherits a monad structure.

[1] Throughout, we restrict to connected migration topologies ($\lambda_2 > 0$). The disconnected case ($\lambda_2 = 0$) reduces to n independent runs where genetic drift erodes diversity; this lies outside the island model proper.

$$\begin{tikzcd} P \arrow[r, "\mathsf{eval}_\phi"] \arrow[rrr, bend right, "\pi = \mathsf{mut}_\delta \circ_K \mathsf{cross} \circ_K \mathsf{sel} \circ_K \mathsf{eval}_\phi"'] & P \arrow[r, "\mathsf{sel}"] & P \arrow[r, "\mathsf{cross}"] & P \arrow[r, "\mathsf{mut}_\delta"] & P \end{tikzcd}$$

Figure 1: The single-generation pipeline in the Kleisli category $\mathbf{Kl}(T)$, where $P = \mathsf{Pop}(\Sigma)$. Each arrow is a Kleisli morphism $P \to TP$ and the curved arrow is their composite π—itself a Kleisli morphism.

2.2 Operators as Kleisli Morphisms

Definition 3 (Genetic operator)**.** *A* genetic operator *is a Kleisli morphism for T: an arrow $f : \mathsf{Pop}(A) \to T(\mathsf{Pop}(B))$, where A and B are genome types (possibly equal). We write $\mathbf{Kl}(T)$ for the Kleisli category whose objects are population types and whose morphisms are genetic operators.*

The standard GA operators are Kleisli morphisms:

- **Evaluation.** $\mathsf{eval}_\phi : \mathsf{Pop}(\Sigma) \to T(\mathsf{Pop}(\Sigma))$, parameterized by a fitness function $\phi : \Sigma \to \mathbb{R}$. Pairs each genome with its fitness.
- **Selection.** $\mathsf{sel} : \mathsf{Pop}(\Sigma) \to T(\mathsf{Pop}(\Sigma))$—an endomorphism projecting the population onto a fitter subset (consuming PRNG state, logging diversity).
- **Crossover.** $\mathsf{cross} : \mathsf{Pop}(\Sigma) \to T(\mathsf{Pop}(\Sigma))$—recombines genetic material from pairs of individuals at a random point, with probability read from Γ.
- **Mutation.** $\mathsf{mut}_\delta : \mathsf{Pop}(\Sigma) \to T(\mathsf{Pop}(\Sigma))$—an endomorphism parameterized by $\delta : \Sigma \to T(\Sigma)$ (Gaussian noise for $\mathbb{R}^n$, bit-flip for $\{0,1\}^k$).
- **Migration.** A natural transformation $\prod_{i \in I} \mathsf{Pop}(\Sigma) \to \prod_{i \in I} T(\mathsf{Pop}(\Sigma))$ transporting individuals between subpopulations without inspecting them. Its naturality underlies the strict/lax dichotomy.

Operators compose via standard Kleisli composition [16]. Figure 1 illustrates the pipeline.

2.3 Island-Model Composition and the Laxator

The Kleisli category supports composition at three levels: operators compose into pipelines (one generation), pipelines compose into strategies (multi-phase evolution), and strategies compose into multi-population systems via the island functor.

A *pipeline* (one generation) is the Kleisli composite $\pi = \mathsf{mut}_\delta \circ_K \mathsf{cross} \circ_K \mathsf{sel} \circ_K \mathsf{eval}_\phi$. Running n generations is the n-fold self-composition π^n. A *strategy* composes pipelines with different parameters, e.g., an exploration phase followed by a convergence phase.

The key construction is the *island functor* $\mathscr{I}_G : \mathbf{Kl}(T) \to \mathbf{Kl}(T^n)$, which lifts a single-population strategy to an n-island system with migration governed by graph G. Here T^n is the n-island monad: each island carries its own PRNG state S_i, logs to its own Λ_i, and shares a common configuration Γ. Concretely, $T^n A = \Gamma \to (S_1 \times \cdots \times S_n) \to (A^n \times (S_1 \times \cdots \times S_n) \times (\Lambda_1 \times \cdots \times \Lambda_n))$. On objects, $\mathscr{I}_G(\mathsf{Pop}(\Sigma)) = \mathsf{Pop}(\Sigma)^n$. On morphisms, $\mathscr{I}_G(\sigma)(P_1, \ldots, P_n)$ applies σ to each island independently, then redistributes individuals along edges of G. Whether $\mathscr{I}_G$ preserves composition depends on G: when G has no edges, composition is preserved exactly (strict functor); when G has edges, migration after each composed step differs from migration after their composition—the discrepancy is precisely the laxator.

Definition 4 (Laxator)**.** *Let $\mathscr{I}_G$ be the island functor for migration graph G. The* laxator *of $\mathscr{I}_G$ is the natural transformation*

$$\varphi_G \;:\; \mathscr{I}_G(\sigma_1) \circ_K \mathscr{I}_G(\sigma_2) \;\Rightarrow\; \mathscr{I}_G(\sigma_1 \circ_K \sigma_2),$$

measuring the discrepancy between composing island-level strategies and lifting a composed strategy. When G has no edges, $\varphi_G = \mathrm{id}$ and $\mathscr{I}_G$ is a strict monoidal functor. Whenever G is connected, $\varphi_G \neq \mathrm{id}$, making $\mathscr{I}_G$ lax.

Remark 1 (Status of the laxator)**.** *The explicit construction of φ_G—giving its components and verifying coherence conditions—is the central open problem of this framework. In this paper, we take a programmatic approach: we identify the laxator as the key quantity, predict its behavior via spectral theory (Section 3), and confirm the prediction empirically (Section 4). The construction itself is future work.*

3 The Spectral Bridge

The categorical framework identifies the laxator φ_G as the quantity governing diversity dynamics. We now connect its magnitude to a computable graph invariant: the algebraic connectivity $\lambda_2(G)$.

3.1 Algebraic Connectivity and the Diversity Ordering

Conjecture 1 (Spectral prediction of topology ordering)**.** *Let G be a connected migration graph on n vertices. The laxator magnitude $\|\varphi_G\|$ grows with $\lambda_2(G)$. Since larger λ_2 implies faster mixing and greater diversity erosion, the diversity ordering for connected topologies is determined by λ_2: smaller λ_2 yields higher sustained diversity.*

Heuristic argument. The spectral gap γ of the lazy random walk on G satisfies $\gamma \geq \lambda_2(L)/(2\Delta)$ for maximum degree Δ [13], and γ is monotone in λ_2. The mixing time satisfies $t_{\mathrm{mix}} = \Theta(\log(n/\varepsilon)/\gamma)$: smaller λ_2 means slower mixing and higher sustained diversity. Since the laxator measures the discrepancy introduced by migration, and migration's homogenizing effect scales with mixing rate, we expect $\|\varphi_G\|$ to grow with λ_2. A formal proof requires the explicit construction of φ_G (Remark 1); for now, this conjecture is supported by the heuristic above and confirmed empirically in Section 4.

The spectral values for our four connected topologies are:

- Ring C_n: $\lambda_2 = 2(1-\cos(2\pi/n)) \approx 4\pi^2/n^2$
- Star $K_{1,n-1}$: $\lambda_2 = 1$
- Fully connected K_n: $\lambda_2 = n$

For random graphs, λ_2 depends on the realization. Among connected topologies, the predicted ordering from smallest to largest λ_2 (hence highest to lowest diversity) is: ring $>$ star $>$ random $>$ FC for $n \geq 7$.

3.2 The n=5 Boundary

At n=5, $\lambda_2(C_5) = 2(1-\cos(2\pi/5)) = 1.382 > 1.0 = \lambda_2(K_{1,4})$. Taken at face value, λ_2 alone predicts star should preserve slightly more diversity than ring. Empirically, ring edges out star in all six domains, but the difference is not significant ($p = 0.14$, Section 4.5). We interpret this as a *scope condition*: when λ_2 values are close (here differing by only 0.38), second-order effects—notably degree heterogeneity (the star hub sees $n-1$ edges while ring vertices see 2)—may dominate. The robust prediction is that ring and star are *hard to distinguish* at n=5; the λ_2 ordering becomes reliable only when the spectral gap between topologies is large.

3.3 The n=7 Prediction

At n=7, $\lambda_2(C_7) = 2(1-\cos(2\pi/7)) = 0.753 < 1.0 = \lambda_2(K_{1,6})$. The inequality reverses: ring now couples *less strongly* than star. The theory makes a falsifiable prediction: at $n \geq 7$, ring preserves strictly more diversity than star. This is confirmed in Section 4.5.

3.4 Categorical Interpretation

The laxator magnitude grows with λ_2: smaller λ_2 means smaller laxator, closer to strict composition, more diversity preserved. Among connected k-regular graphs, a *Ramanujan* graph maximizes λ_2 (Alon–Boppana bound)—maximally lax. The diversity-optimal (minimally lax) connected topology minimizes λ_2; for k=2 this is C_n, the ring.

The strict/lax dichotomy is thus a continuous family indexed by λ_2, confirmed independently in oscillator synchronization [19].

4 Experiments

4.1 Setup

All experiments: 5 islands $\times$ 16 individuals, 100 generations, migration rate 0.1 every 5 generations, 30 seeds per (topology, domain) pair. We sweep five topologies: none (no migration), ring, star, random (Erdős–Rényi, resampled each migration event), and fully connected. Diversity is measured as normalized pairwise Hamming distance (binary genomes) or Euclidean distance (real-valued genomes), consistent with Leinster's axiomatic framework [12].

4.2 Six Domains

We select six domains spanning three axes: landscape structure (monotone vs. intransitive), representation (fixed-length vs. variable-length), and constraint type (unconstrained, hard, soft). The categorical structure—monad stack, pipeline shape, composition operator—is identical across all domains; only genome type and fitness function change.

OneMax [15]: 20-bit bitstrings, fitness = number of ones. The simplest possible landscape. **Maze generation**: 60-bit binary vectors encoding wall presence on a 6×6 grid; fitness combines BFS-derived path length, dead-end ratio, and branching factor. **Graph coloring**: 3-coloring of random graphs; fitness penalizes constraint violations. **Knapsack**: classic NP-hard optimization with capacity constraints. **No Thanks!**: co-evolutionary card game; fitness is relative to opponents within each island. **Checkers**: 64-bit binary genomes encoding board-evaluation weights; fitness determined by round-robin tournament within each island, creating an intransitive landscape where fitness depends on co-evolving opponents [18].

4.3 Main Result: Universal Topology Ordering

Observation 1 (Domain-independent topology ordering)**.** *Across six unrelated domains, the diversity ordering at generation 99 is none > ring > star > random > FC with perfect concordance: Kendall's* $W = 1.0$*,* $\chi^2 = 24.0$*,* $p = 0.00008$*; all 15 pairwise Spearman* $\rho = 1.0$ *(Figure 2). A two-way ANOVA confirms topology explains* $28.7\times$ *more variance than domain.*

The none-to-ring transition produces a 35% diversity drop—the symmetry-breaking first coupling dominates. Each subsequent step contributes at most 9%. The laxator norm is continuous but non-uniform: the first coupling is qualitatively different from subsequent refinements.

Sorting network was tested as a seventh domain but is degenerate: all topologies converge identically, identifying a scope condition (when the fitness landscape imposes uniform convergence pressure, topology effects vanish).

4.4 Checkers: Co-evolutionary Stress Test

Checkers is the most demanding test of our invariance claim. Fitness is not a property of a genome but of a genome *relative to its co-evolving opponents*—strategy A may beat B, B may beat C, yet C may beat A. Migration introduces not just genetic material but new competitive context. Despite this, the ordering holds: checkers shows the smallest phase transition (11.1% none-to-ring drop, consistent with co-evolutionary buffering where migrant fitness depends on local context) yet preserves the full five-topology ordering (ring vs. star: Cohen's $d = 0.577$, $p = 0.029$; none vs. FC: $d = 5.76$).

4.5 Spectral Predictions Confirmed

The n=5 boundary. At n=5, Conjecture 1 predicts ring and star should be hard to distinguish ($\lambda_2(C_5) = 1.382$ vs. $\lambda_2(K_{1,4}) = 1.0$, a gap of only 0.38). Confirmed: Fisher's combined $p = 0.14$ across six domains—ring and star diversities are statistically indistinguishable. Note that ring edges out star consistently despite having higher λ_2; we attribute this to degree-heterogeneity effects that dominate when λ_2 values are close (Section 3).

The n=7 confirmation. At n=7, the spectral bridge predicts ring > star in diversity ($\lambda_2(C_7) = 0.753 < 1.0 = \lambda_2(K_{1,6})$). A confirmatory maze experiment (30 seeds, identical parameters) is unambiguous: ring diversity 0.387 ± 0.028 versus star 0.336 ± 0.053 ($p = 6.6 \times 10^{-5}$). Pure graph theory, oscillator physics [19], evolutionary graph theory [4], and our experiments converge—not just on the ordering, but on the *boundary* where it changes.

Time-varying topology and the $5.5\times$ inflation. For fixed graphs, snapshot $\lambda_2(G)$ perfectly predicts the diversity ordering. For the time-varying random topology (resampled each migration event), snapshot λ_2 fails: $\bar{\lambda}_2 = 0.36$ is the lowest of all connected topologies, yet random ranks fourth. The time-averaged adjacency matrix yields effective $\lambda_2 = 39.74$ ($110\times$ the snapshot mean), recovering the ordering. Categorically, fixed topology makes $\mathscr{I}_G$ a strict endofunctor; time-varying topology produces a genuinely lax endofunctor whose $5.5\times$ per-event diversity inflation is the laxator's numerical signature.

4.6 Fingerprints: Composition Determines Trajectory Shape

Definition 5 (Diversity fingerprint)**.** *Let $\sigma : \mathsf{Pop}(\Sigma) \to T(\mathsf{Pop}(\Sigma))$ be a strategy (a composed Kleisli morphism). The* diversity fingerprint *of σ is the sequence $\mathbf{d}(\sigma) = (d_0, d_1, \ldots, d_n) \in \mathbb{R}^{n+1}$, where d_i is the genotypic diversity at generation i.*

Beyond the topology ordering, we observe that strategies with identical composition patterns produce matching diversity trajectory *shapes* across all tested domains: flat (monotonic decline), hourglass (spike–crash–rebound), and island (stable oscillation maintained by migration). The categorical explanation: fingerprints are functions of the composition pattern, which is invariant across domains. A formal

definition of "shape equivalence" remains open, but the empirical pattern is consistent across all six domains.

5 Discussion

Naturality interpretation. Why is the ordering domain-independent? Because it depends on *compositional structure*—preserved by the domain-change functor—not on *content* (genome type, fitness function). $W = 1.0$ across six domains is *consistent with* the topology ordering being a natural transformation: when we replace checkers genomes with maze genomes, the ordering is unchanged, just as a natural transformation commutes with the underlying functor. Making this precise requires formalizing the domain-change functor and showing the ordering commutes with it—a goal for future work.

Practical implications. For EC practitioners: topology choice matters more than domain-specific tuning. Ring is optimal among connected topologies. At $n \geq 7$ islands, ring clearly outperforms star; at n=5, they are interchangeable. The 28.7× variance ratio suggests that investing effort in topology selection yields greater returns than investing in domain-specific diversity-preservation mechanisms.

Connections to other fields. Four independent lines confirm the λ_2 universality: λ_2 governs synchronization onset in coupled oscillators [19], consensus time scales as N^3 on cycles versus N^2 on complete graphs [4], threshold graphs are globally synchronizing [22], and local structure determines topologically protected synchronization [17]. Li et al. [14] formalize emergence via derived functors, paralleling the laxator from a homological algebra perspective. Ghani et al. [7] show that exact equilibria compose cleanly but approximate ones do not—the game-theoretic analogue of our strict/lax dichotomy.

The general conjecture. We conjecture this pattern extends beyond EC: across optimization paradigms, strict composition preserves compositional invariants while lax composition degrades them [5, 11, 8, 1, 2, 19, 9].

Per-island asymmetry. At intermediate topologies (star, random), graph heterogeneity produces per-island synchronization asymmetry [24]; full time series are in the companion repository.

Limitations. (a) The laxator is identified but not explicitly constructed (Remark 1)—the central open problem. (b) Experiments are at n=5 and n=7; larger n is untested. (c) Sorting network is degenerate, identifying a scope condition: domains with uniform convergence pressure may suppress topology effects. (d) All experiments use homogeneous islands (same strategy on each island). (e) At n=5, λ_2 predicts star > ring, but ring edges out star consistently; degree heterogeneity likely introduces a correction term not captured by λ_2 alone.

Future work. Three directions: (a) explicit construction of the laxator φ_G and verification of coherence conditions; (b) heterogeneous islands with braided monoidal migration; and (c) scaling to larger n to map the full λ_2 landscape.

6 Conclusion

Migration topology determines diversity dynamics independently of fitness landscape. The ordering none > ring > star > random > fully connected holds with perfect rank correlation across six fundamentally different domains (Kendall's $W = 1.0$, $p = 0.00008$), with topology explaining $28.7\times$ more variance than domain.

The mechanism: a categorical framework identifies GA operators as Kleisli morphisms over a population monad, with the island functor's laxator measuring migration-induced coupling. A spectral bridge connects the laxator to $\lambda_2(G)$, predicting the diversity ordering. The prediction is confirmed at n=5 (ring/star indistinguishable, $p = 0.14$) and n=7 (ring clearly outperforms star, $p = 6.6 \times 10^{-5}$)—the theory predicts not just the ordering but the boundary where it changes.

Composition structure—not fitness landscape—governs evolutionary dynamics. The explicit construction of the laxator, heterogeneous islands, and formal proof of the general conjecture are the next steps toward a complete categorical theory of evolutionary computation.

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../experiments/plots/multi_domain_topology_ordering.pdf

Figure 2: Diversity (mean $\pm$ SE, 30 seeds, generation 99) across all six experimental domains: OneMax, Maze, Graph Coloring, Knapsack, No Thanks!, and Checkers. The canonical ordering none $>$ ring $>$ star $>$ random $>$ FC is identical in every domain (Kendall's $W = 1.0$, $p = 0.00008$). Topology explains $28.7\times$ more variance than domain (two-way ANOVA).